TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	82.6 ± 1.50
Correlation Stability	80.3 ± 2.53
Distribution Consistency	90.0 ± 1.95
Entropy Preservation	93.5 ± 0.92
Mean Z Score Drift	0.1 ± 0.02
Categorical Fidelity	91.9 ± 1.58
Privacy Risk	51.7 ± 1.35
Data Density	100.0 ± 0.01
Coverage Score	17.2 ± 3.09
Feature Accuracy	88.9 ± 3.53
Novelty Score	0.4 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	claude-sonnet-4-6
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	48.753	+4.007	9.0%	OK
TB	3.299	3.138	-0.161	4.9%	OK
DB	1.486	1.446	-0.040	2.7%	OK
Alkphos	290.576	271.009	-19.568	6.7%	OK
Sgpt	80.714	51.189	-29.525	36.6%	Drift
Sgot	109.911	67.594	-42.316	38.5%	Drift
TP	6.483	6.344	-0.139	2.1%	OK
ALB	3.142	3.069	-0.073	2.3%	OK
A/G Ratio	0.947	0.930	-0.017	1.8%	OK
Selector	1.286	1.416	+0.130	10.1%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	82.6 ± 1.50
Correlation Stability (*2)	80.3 ± 2.53
Distribution Consistency (*3)	90.0 ± 1.95
Entropy Preservation (*4)	93.5 ± 0.92
Mean Z Score Drift (*5)	0.1 ± 0.02
Categorical Fidelity (*6)	91.9 ± 1.58
Privacy Risk (*7)	51.7 ± 1.35
Data Density (*8)	100.0 ± 0.01
Coverage Score (*9)	17.2 ± 3.09

Feature Accuracy (*10)	88.9 ± 3.53
Novelty Score (*11)	0.4 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6777777777777778	0.7025641025641025	0	
F1	0.7997903571549948	0.762375061172731	0.7939166859994153	-0.005873671155579341	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.7042735042735042	0.6914529914529914	-0.014529914529914534	
F1	0.8128002697439015	0.779896703720204	0.783757490909639	-0.029042778834262484	*

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743589	0.6829059829059829	0.6128205128205129	0.05384615384615384	*
F1	0.5587432119380917	0.751837948478304	0.6424389532560584	0.0836957413179667	*

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6752136752136753	0.6940170940170941	-0.01282051282051282	
F1	0.8279091624529537	0.7368803812572882	0.7746903449562248	-0.05321881749672896	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6547008547008548	0.6170940170940171	0.6606837606837607	0.005982905982905984	
F1	0.7597542392611762	0.6937378291216186	0.7558708663853764	-0.0038833728758000618	

Model: KNN

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6709401709401713	0.6863247863247863	-1.1102230246251566e-17	
F1	0.7971299319768493	0.7398311095235237	0.7765755176402708	-0.020554414336578463	*

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6820512820512821	0.6632478632478632	0.6948717948717948	0.01282051282051282	
F1	0.7881992596107154	0.7457797115424644	0.788519738720846	0.00032047911013091126	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6769230769230773	0.6888888888888889	0.014529914529914534	
F1	0.7821343746683586	0.7600537592899006	0.7817120605941665	-0.00042231407419216 49	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.688034188034188	0.6897435897435897	0.6820512820512821	-0.005982905982905984	
F1	0.796509885617606	0.766057437617164	0.7661125654987591	-0.030397320118846837	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.705982905982906	0.6871794871794872	0.7196581196581198	0.013675213675213682	
F1	0.8096218881259961	0.764970799656288	0.8083053131985471	-0.0013165749274491767	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
2	37	Male	1.3	0.4	195	41	38	5.3	2.1	0.6
3	60	Male	0.7	0.2	174	32	14	7.8	4.2	1.1
4	42	Female	0.8	0.2	195	18	15	6.7	3.0	0.8
5	51	Male	2.9	1.3	482	22	34	7.0	2.4	0.5
6	42	Male	0.8	0.2	198	29	19	6.6	3.0	0.8
7	45	Male	1.2	0.4	210	35	42	6.8	3.2	0.9
8	33	Female	0.7	0.2	172	22	18	7.1	3.9	1.2
9	58	Male	3.4	1.6	312	68	95	5.6	2.3	0.6
10	29	Female	0.8	0.2	165	19	14	7.3	4.0	1.2
11	61	Male	9.5	4.8	420	85	110	5.2	2.0	0.6
12	43	Male	0.9	0.3	185	31	27	6.6	3.3	1.0
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	1.7	0.8	331	36	53	7.3	3.4	0.9
2	52	Male	0.8	0.2	245	48	49	6.4	3.2	1.0
3	42	Female	0.8	0.2	182	22	20	7.2	3.9	1.1
4	45	Female	3.5	1.5	189	63	87	5.6	2.9	1.0
5	38	Female	0.8	0.2	185	25	21	7.0	3.0	0.7
6	45	Male	2.2	0.8	209	25	20	8.0	4.0	1.0
7	44	Male	1.4	0.6	210	38	45	7.1	3.6	1.0
8	53	Female	0.8	0.2	195	22	18	6.8	3.4	1.0
9	37	Male	2.8	1.3	275	55	72	6.3	2.9	0.8
10	61	Male	5.2	2.6	380	88	110	5.8	2.3	0.6
11	29	Female	0.7	0.2	155	19	22	7.4	4.1	1.2
12	48	Male	3.1	1.4	318	67	85	6.6	3.0	0.8
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
2	65	Male	1.4	0.6	260	28	24	5.2	2.2	0.7
3	68	Male	0.7	0.1	145	20	22	5.8	2.9	1.0
4	55	Male	0.9	0.2	116	36	16	6.2	3.2	1.0
5	40	Male	0.7	0.2	176	28	43	5.3	2.4	0.8
6	45	Male	2.9	1.4	210	74	68	7.2	3.6	1.0
7	44	Male	1.2	0.4	224	38	42	6.8	3.4	1.0
8	57	Female	0.8	0.2	310	44	51	7.1	3.3	0.9
9	29	Male	0.7	0.1	158	32	28	7.4	3.9	1.1
10	63	Male	4.5	2.2	415	55	88	5.8	2.5	0.7
11	39	Female	0.9	0.3	195	19	23	6.5	3.1	0.9
12	52	Male	2.8	1.3	370	68	95	6.2	2.7	0.8
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
2	12	Male	1.0	0.2	719	157	108	7.2	3.7	1.0
3	35	Female	0.6	0.2	180	12	15	5.2	2.7	nan
4	53	Female	0.9	0.2	210	35	32	8.0	3.9	0.9
5	18	Male	1.8	0.7	178	35	36	6.8	3.6	1.1
6	58	Male	0.9	0.2	1100	25	36	7.1	3.5	0.9
7	34	Male	1.1	0.4	210	45	38	7.1	3.8	1.15
8	52	Female	0.8	0.2	175	28	22	6.9	3.5	1.0
9	43	Male	3.2	1.5	320	110	145	6.5	3.2	1.0
10	29	Female	0.7	0.2	162	18	20	7.3	4.0	1.2
11	61	Male	2.0	0.9	415	55	72	5.8	2.8	0.9
12	38	Male	0.9	0.2	195	40	35	7.6	4.1	1.2
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Female	0.8	0.2	168	25	18	6.2	3.1	1.0
2	40	Male	0.6	0.1	171	20	17	5.4	2.5	0.8
3	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
4	50	Male	0.8	0.2	152	29	30	7.4	4.1	1.3
5	47	Female	0.8	0.2	236	10	13	6.7	2.9	0.76
6	32	Male	12.7	8.4	190	28	47	5.4	2.6	0.9
7	55	Male	1.2	0.5	210	38	45	6.8	3.3	0.9
8	43	Female	0.8	0.2	185	22	20	7.1	3.8	1.1
9	67	Male	3.5	1.8	320	55	88	5.9	2.4	0.6
10	29	Male	0.6	0.1	160	42	35	7.8	4.2	1.2
11	52	Female	1.4	0.6	245	30	28	6.5	3.0	0.8
12	38	Male	8.2	4.1	275	95	180	6.2	2.7	0.7
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
2	72	Male	0.7	0.1	196	20	35	5.8	2.0	0.5
3	56	Male	1.0	0.3	195	22	28	5.8	2.6	0.8
4	62	Male	1.2	0.4	195	38	54	6.3	3.8	1.5
5	17	Male	0.9	0.2	279	40	46	7.3	4.0	1.2
6	18	Male	0.9	0.3	300	30	48	8.0	4.0	1.0
7	45	Male	1.8	0.7	245	55	62	6.9	3.2	0.9
8	38	Female	0.8	0.2	178	22	19	7.1	3.8	1.2
9	52	Male	3.2	1.4	312	88	145	6.3	2.7	0.7
10	29	Male	0.7	0.2	201	31	27	7.5	4.1	1.2
11	61	Male	4.1	1.9	420	65	98	5.8	2.2	0.6
12	44	Female	1.1	0.4	265	29	34	6.6	3.3	1.0
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	78	Male	1.0	0.3	152	28	70	6.3	3.1	0.9
2	38	Male	0.8	0.2	208	25	50	7.1	3.7	1.0
3	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
4	25	Female	0.9	0.3	159	24	25	6.9	4.4	1.7
5	62	Male	1.2	0.4	195	38	54	6.3	3.8	1.5
6	68	Male	0.7	0.1	145	20	22	5.8	2.9	1.0
7	54	Male	1.1	0.4	210	35	45	6.8	3.4	1.0
8	43	Female	0.8	0.2	165	22	18	7.0	3.6	1.1
9	67	Male	3.2	1.4	320	55	80	6.1	2.8	0.8
10	29	Male	0.7	0.2	180	40	32	7.2	3.9	1.2
11	51	Female	1.5	0.6	275	48	60	6.4	3.2	1.0
12	36	Male	0.9	0.3	190	30	27	7.5	4.0	1.1
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	55	Male	0.8	0.2	290	139	87	7.0	3.0	0.7
2	38	Male	3.7	2.2	216	179	232	7.8	4.5	1.3
3	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	58	Male	0.8	0.2	123	56	48	6.0	3.0	1.0
6	34	Male	6.2	3.0	240	1680	850	7.2	4.0	1.2
7	45	Male	1.2	0.4	210	55	72	6.8	3.4	1.0
8	38	Female	0.9	0.2	185	28	35	7.2	3.8	1.1
9	62	Male	4.1	2.0	320	88	145	5.8	2.6	0.8
10	29	Male	0.7	0.1	165	32	28	7.9	4.1	1.1
11	55	Female	2.8	1.3	275	48	62	6.1	2.8	0.8
12	41	Male	1.5	0.6	230	75	98	6.5	3.2	0.97
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
2	38	Female	0.8	0.2	185	25	21	7.0	3.0	0.7
3	75	Male	0.9	0.2	282	25	23	4.4	2.2	1.0
4	60	Male	6.8	3.2	308	404	794	6.8	3.0	0.7
5	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51
6	50	Male	2.6	1.2	415	407	576	6.4	3.2	1.0
7	43	Male	1.2	0.4	220	45	52	6.8	3.3	1.0
8	55	Female	0.8	0.2	195	28	24	7.1	3.8	1.2
9	38	Male	4.5	2.1	375	185	210	6.5	3.0	0.8
10	62	Male	0.7	0.2	178	32	38	5.9	2.6	0.8
11	29	Female	0.9	0.3	165	19	17	7.4	4.0	1.2
12	47	Male	8.2	3.9	520	98	175	6.2	2.8	0.8
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	36	Female	0.8	0.2	650	70	138	6.6	3.1	0.8
2	40	Male	0.6	0.1	171	20	17	5.4	2.5	0.8
3	31	Female	1.1	0.3	190	26	15	7.9	3.8	0.9
4	32	Male	0.7	0.2	189	22	43	7.4	3.1	0.7
5	46	Male	3.3	1.5	172	25	41	5.6	2.4	0.7
6	4	Male	0.8	0.2	460	152	231	6.5	3.2	0.9
7	43	Male	1.2	0.4	210	35	42	6.8	3.2	0.9
8	55	Female	0.7	0.2	175	22	19	6.5	3.4	1.1
9	38	Male	2.4	1.1	265	58	87	6.1	2.8	0.8
10	61	Male	0.9	0.3	188	31	28	6.3	3.1	1.0
11	29	Female	0.8	0.2	195	27	22	7.6	3.7	0.95
12	50	Male	3.7	1.8	310	47	63	5.8	2.5	0.7
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3